PAPER TRAIL

Progress Report: changes since the 30 September 2026 project report

Period covered	1 October 2026 to 10 October 2026 (all code changes landed on 9 October)
Baseline	Project report dated 30 September 2026 (repository commit 9dcaa22)
Branch	feature/claim-accuracy (9 commits, 44 files changed, +4,166 / −473 lines)
Scope	Only what changed. Models, datasets, fusion model, thresholds and the article-path benchmarks are unchanged and are not repeated here.

1. Summary

The previous report ended with an article-verification pipeline that was accurate on whole articles but unreliable on the short, one-sentence inputs that people actually type into the verify page, and that depended on a human review queue for uncertain cases. Since then the work has focused on **claim accuracy**: making the system give a correct, evidence-based answer for short claims and breaking-news style claims, and making it aware of *where* a claim is about.

21% → 97%	**0 / 15**	**123 → 259**	**1 new stage**
accuracy on committed verdicts, 40 never-tuned single claims	fabricated breaking-news variants called real	automated test functions	locate (where is the claim about?)

- **Human review removed.** The review queue, its table and its endpoints are gone; the system always answers on its own, returning `uncertain` with the reason attached.
- **Region-aware verification.** A new `locate` stage finds the place a claim is about (street up to country) and adds regional news to the evidence.
- **Claim mode.** One- or two-sentence inputs are decided on evidence alone, because the style classifier is out of distribution on single sentences.
- **Breaking-news handling.** Time-bound claims can no longer be refuted by other days' news, and can be confirmed by several recent independent reports.
- **Indian-source weighting.** Indian outlets now outrank foreign ones, and Wikipedia no longer leads live-news claims.

2. What changed at a glance

Change	Commit	Effect
Remove human review	a32c884	Review-queue router, `ReviewQueue` table and `needs_human_review` flag deleted; 20 files, +47 / −393.
Region-aware claim verification	0a8ee77	New locator, gazetteer and geocoder modules; regional news retrieval; new response fields; frontend shows detected place. 22 files, +2,599.
Locator bug fixes	4b3e7b7	Six defects found by measuring the locator on a labelled sample, with new tests.
Claim mode	31235cb, 7e3c13c	Evidence-only verdicts for short inputs; compound-claim splitting; subject-only refutation. Documented with measured results.
Time-bound claims cannot be refuted by other days' news	1a93691	Fixes a real breaking-news headline being judged `likely_fake`.
Event corroboration	8fd8370	New `event_match` module; recent independent reports can confirm a time-bound claim at capped confidence.
Indian-source weighting	09240ef	Source trust tiers re-ordered; stale cached trust fixed; honest abstention display.
Dependency lockfile	a6e0154	`package-lock.json` added.

3. Changes in detail

3.1 Human review removed from the pipeline and API

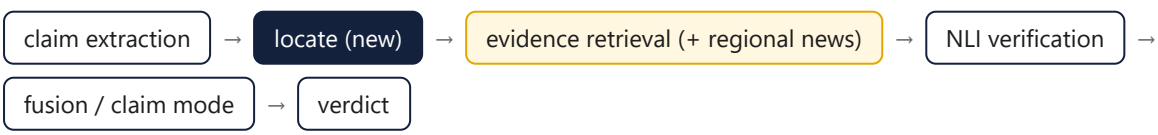
What was done. The `/review-queue` endpoints, the `ReviewQueue` table and the `needs_human_review` flag were deleted from the code. Failures and low-confidence cases now return `uncertain`, with the reason written into `reasoning_trace` and `degradations`. The old `verdicts.needs_human_review` column is left in existing databases (unused) so earlier databases keep accepting inserts.

Unchanged. Fusion model, thresholds and trained models. The abstention band still decides when to answer; only the destination of an abstained article changed.

3.2 Region-aware claim verification

Problem. Many claims are about a specific place, and the best evidence for them is local news, which global search and wire services often miss.

What was built.



Dark box: new stage. Amber box: existing stage extended. Review step removed (3.1).

- **Locator** (`agents/locator.py`, `nlp/gazetteer.py`, `nlp/geocoder.py`). Resolves each claim's place as precisely as the text supports (site, street, locality, city, district, state, country) with a confidence, using a

built-in gazetteer plus OpenStreetMap Nominatim (free, cached on disk, rate-limited to one request per second).

- **Refuses to guess.** Ambiguous names such as "Springfield" are reported as not located instead of guessed; places below 0.5 confidence are dropped.
- **Regional retrieval.** For a located claim, regional news (Google News RSS country editions, or newsdata.io with a free key) is added *after* the existing global search and widened locality → city → state → country until enough relevant snippets are found. Global results are untouched and stay first.
- **Controls.** `REGION_AWARE` switches the feature off entirely; submit endpoints accept an optional `region` hint.

> **Honest limit.** On the encyclopedic claims used for measurement, region-aware search made no difference (identical results on and off). Its value is for local news claims, which have no benchmark yet.

3.3 Locator defects fixed after measurement

The locator was run against a labelled sample and six defects were fixed (with new tests in `tests/test_locator.py`):

- "New South Wales" no longer resolves to Wales.
- An institution followed by a verb such as "announced" is no longer mistaken for a person.
- Places renamed in OpenStreetMap match by their old names (Aurangabad).
- "City, State" and "X in Y" give the geocoder a qualifier.
- Place cues are judged around the trimmed name ("Residents of Whitefield").
- Kashmir resolves to Jammu and Kashmir through a gazetteer alias.

3.4 Claim mode: short inputs decided on evidence alone

Problem. The stylistic ensemble and the learned fusion were fitted on whole articles. On a single sentence the classifier is out of distribution (it said "real" at 0.99 about plainly false sentences) and the fusion followed it. Measured on 40 fresh single claims, the article path committed to a verdict with only **21% accuracy**, worse than guessing.

What was built. Inputs of at most 300 characters and at most 2 sentences take a separate path (`CLAIM_MODE=false` disables it; longer text is unchanged):

Evidence result	Claim-mode verdict
Supported (entailment mass ≥ 0.45)	`likely_real`
Refuted (contradiction mass ≥ 0.5) and nothing supports it	`likely_fake`
Inconclusive, mixed or no evidence	`uncertain` (never `likely_fake`)

- **Compound claims are split** ("X is A and is B" is checked as two claims).
- **Each named subject is looked up directly**, and a Wikipedia page titled in the claim counts as on-topic even when the claim gets a detail wrong.
- **Only a page about the claim's own subject, or a near-paraphrase, may refute it.** Neighbouring pages (Blue Nile for a claim about the Nile, Agra Fort for the Red Fort) caused false refutations and may now only support.
- **Very short claims are accepted** (10 characters and up), and short plain-ASCII text is no longer rejected as an "unsupported language" (a 48-character English sentence had been labelled Catalan at 0.99999).

3.5 Breaking news: time-bound claims

Refutation guard (1a93691). A real breaking-news headline was judged `likely_fake` because regional news about the same story (17 stations closed on day 3) was read as contradicting the claim (57 stations on day 1). Time-bound claims (weekday, clock time, month and day, "live updates", "ahead of") and regional news items can now *support* a claim but not *refute* it.

Event corroboration (8fd8370). A breaking story is often restated with changed figures the next day, so NLI finds no support. When at least 3 news items from at least 2 outlets, dated within 7 days, cover the claim's distinctive terms (acronyms must match), the claim is `likely_real` at a capped confidence of 0.65, and is flagged when its figures differ (`detail_mismatch`). Reference pages do not count. Tested against fabricated variants: **0 of 15 were called real**.

3.6 Indian-source weighting

- **Trust tiers.** Indian outlets 0.82 to 1.0, unlisted `.in` domains 0.65, foreign news scaled by `FOREIGN_NEWS_FACTOR` (0.7), Wikipedia 0.6. NewsAPI now queries the Indian domains too.
- **Retrieval.** Cached evidence is re-stamped with the current trust table on every read (cached entries had kept stale weights); global and regional evidence are merged best-weighted first; for time-bound claims one Wikipedia fallback is kept once two news items exist; empty keyless NewsAPI results are no longer cached.
- **Display.** An abstention in claim mode now has no confidence value instead of a fake 50%, and the verify page reads "Not enough evidence to decide".

4. API, database and interface changes

Area	Change
Endpoints removed	`GET /review-queue` and `POST /review-queue/{id}/resolve`. Demo users `admin` and `analyst` now have the same access.
Request	`POST /articles/submit` and `/articles/submit-image` accept an optional `region` (up to 120 characters).
Response: `claim_verdicts[]`	New fields: `location` (name, label, level, coordinates, hierarchy, confidence, source), `search_scope`, `regional_evidence`, `in_region`, `event_corroborated`, `corroborating_sources`, `detail_mismatch`. All optional, absent on older rows.
Database	`review_queue` table and `verdicts.needs_human_review` removed from the ORM; existing databases keep the unused table and column.
Frontend	Verify page shows the detected place per claim, the corroboration note and the "Not enough evidence to decide" state.
New settings	`REGION_AWARE`, `GEOCODER_BACKEND`, `REGIONAL_NEWS_PROVIDER`, `REGION_*`, `CLAIM_MODE`, `CLAIM_MODE_*`, `CLAIM_ENTAIL_MIN`, `CLAIM_CONTRADICT_MIN`, `FOREIGN_NEWS_FACTOR`, `EVENT_*` (all documented in `.env.example`).
Documentation	README, `docs/VERIFICATION_PIPELINE.md` (new locate and claim-mode sections) and `docs/CURRENT_STATE.md` updated.

5. Measured results

Hand-written true and false claims about landmarks, places and facts (half true, half false by swapping one detail), with Wikipedia as the main evidence. The "tuned" sets were used while building claim mode; the last set was written beforehand and looked at once.

Set	Path	Committed	Accuracy when committed	Strict accuracy*
40 claims, written before tuning	article path	70%	**21%**	15%
same, after tuning	claim mode	72%	**100%**	72%
24 regional claims (tuned on)	article path	67%	38%	25%
same, after tuning	claim mode	62%	100%	62%
40 fresh claims, never tuned on	article path	75%	**30%**	22%
same	claim mode	92%	**97%** (1 wrong)	90%
Golden set, 30 articles	unchanged	57%	76%	43%

* correct divided by all inputs, counting `uncertain` as not correct.

> **Read these with care.**
>
> - The claims are entity–attribute facts ("X is in Y", "X was built in 1889"), which Wikipedia answers well. They say nothing about fabricated events, quotes, statistics or satire, where the right answer is mostly `uncertain`.
> - The sets are small (24 to 40 claims). 97% on 37 committed verdicts is not a population estimate.
> - The remaining error mode is NLI misreading a true statement as contradicted (for example a true Hampi/Vijayanagara claim at 0.85).
> - The breaking-news result (0 of 15 fabricated variants called real) is a safety check on a small set, not a measure of recall on real breaking news.

6. Unchanged, and still open

Unchanged since 30 September: the 22-model classifier bank and benchmark table, the stacking ensemble, the DeBERTa-v3 NLI model, the 13-feature fusion model and its 0.395 / 0.605 thresholds, the datasets and the article-path evaluation (holdout 98.3% at 42% coverage, golden 88.2% at 57% coverage).

Still open from the previous report:

- Models 1, 2 and 3 (multilingual style, region classifier, fine-tuned claim NLI) are planned for Colab and not built.
- Stacking-ensemble leakage fix (out-of-fold predictions) and expanding the golden set beyond 30 cases.
- No evaluation yet of OCR accuracy or Hindi translation quality.
- No Docker or CI configuration; demo authentication and open CORS still need replacing before production.

New limitations introduced by this work:

- No benchmark yet for local-news or breaking-news claims, so the value of region-aware retrieval and event corroboration is untested beyond spot checks.

- Nominatim is limited to one request per second (mitigated by caching and a per-article budget of 8 places).
- Google News RSS is intended for personal, non-commercial use; `newsdata.io` or `none` should be used beyond a coursework demo.
- Claim mode defers to evidence only, so claims with thin or no public evidence return `uncertain` by design.